

KYC/EDD EVIDENCE ENGINE v8

MVP Evaluation Access and Project Overview

CONFIDENTIAL - CONTAINS ACTIVE ACCESS TOKEN

Evaluation website

<https://kyc-edd-v8-gateway.kycsweep.workers.dev/>

Website evaluation token

83c06eb50e020c34228b7626d1

This token is separate from the password used to open the final PDF.

Access expiry	28 August 2026 at 23:59:59 UTC
Production environment	Cloudflare Workers - workers.dev evaluation deployment
Active deployment	aa40fc1c-3a92-4212-8b02-f46c7f0e2226
Health status at handoff	Verified HTTP 200 with minimal response {"status": "ok"}

Quick evaluation instructions

1. Open the website using the link above.
2. Paste the website evaluation token into the Temporary evaluation access token field.
3. Select Demo A: no candidate to review the deterministic no-candidate workflow.
4. Select Demo B: potential match to review candidate resolution and manual-review handling.
5. Use live screening only with authorised, consented, synthetic, or otherwise lawful evaluation data.
6. Select Forget token after the evaluation and close the browser window.

Demo safety

Demo A and Demo B use deterministic server-authored fixtures. They do not call live external sources and do not create screening records in Cloudflare D1.

1. Evaluation Scope and Data Handling

What the evaluator can assess, how results should be interpreted, and how the MVP handles access and screening data.

Evaluation scope

- Browser-based access through one HTTPS link and one temporary evaluation token.
- Sanctions and watchlist candidate screening through OpenSanctions.
- PEP and related-close-associate candidate screening through OpenSanctions.
- Bounded adverse-media and public-news discovery through the GDELT DOC API.
- Country-risk context for analyst-provided nationality and residence.
- Deterministic ten-section audit output, case identifiers, source status, and SHA-256 report integrity metadata.
- Synthetic no-candidate and potential-match demonstrations.
- PDF and JSON report export from the evaluator interface.

Result semantics

State	Meaning in this MVP
NO_MATCH	No candidate was returned within enabled coverage and configured retrieval thresholds. It is not proof that no relevant record exists.
POTENTIAL_MATCH	A candidate requires analyst-led identity resolution. It does not constitute a confirmed match.
INCONCLUSIVE / SOURCE_ERROR	Coverage was incomplete or a required source failed, timed out, or was rate-limited. The system does not convert this state into a clear result.
NOT_ASSESSED	The section is outside enabled MVP source coverage or lacks sufficient evidence for assessment.

Security and data handling

- The evaluation token is held only in volatile browser-page memory and is sent to the gateway in the X-Evaluation-Token HTTPS header.
- The token and provider API credentials are server-side secrets and are excluded from reports and D1 storage.
- Live screening audit bundles are stored temporarily in Cloudflare D1 with a seven-day purge period.
- Synthetic demonstrations make no live provider request and create no D1 screening record.
- No evaluator-facing route can list, search, export, or retrieve historical D1 screening records in the current MVP.
- The public health endpoint returns only a minimal status response.
- Rate limiting and an exact D1 upstream-usage counter restrict abuse and external-provider consumption.

Responsible-use requirement

Do not enter personal data unless you are authorised to process it for a lawful evaluation purpose. The application is not documentary identity verification and is not a final compliance-decision engine.

2. Project Status and Recognised Limitations

Project status

This submission is a minimum viable product developed within a highly constrained 48-hour delivery period. It demonstrates a bounded KYC/EDD OSINT workflow and the technical integration needed to produce a transparent, reviewable audit report.

Present MVP capabilities

- Sanctions and watchlist candidate screening.
- PEP and related-close-associate candidate screening.
- Bounded adverse-media and public-news discovery.
- Nationality and residence country-risk context.
- Immutable analyst-provided subject baseline.
- Candidate matching with explicit conflicts and manual-review recommendations.
- Source-status, coverage, and failure-state reporting.
- Deterministic audit-bundle generation and SHA-256 report hashing.
- Temporary D1 retention and scheduled deletion of expired screening records.
- Synthetic no-candidate and potential-match test pathways.

Recognised limitations

- Limited external-source coverage and dependence on third-party availability and rate limits.
- Bounded query combinations rather than a complete combinatorial search strategy.
- Narrow English-title claim extraction and no general web-page scraping.
- No documentary identity verification, biometric verification, or proof-of-address validation.
- No evaluator-facing historical case-management interface.
- No complete applicant-level Low, Medium, or High risk classification.
- No automated final compliance determination.
- No individual user accounts, role-based access control, or per-user revocation; access is based on possession of the shared evaluation token.
- The optional Brave Search connector is implemented but disabled in the final evaluation configuration.

Observed evaluation constraints

External sources may return rate-limit or availability errors. The live test demonstrated that OpenSanctions could complete while GDELT returned an HTTP 429 response. The application correctly preserved the source failure and overall inconclusive state rather than presenting an unsupported clear outcome.

3. Recommended Further Development

3.1 Data collection and processing

Re-evaluate the collection and processing mechanism with a deeper, documented combinatorial strategy for:

- legal names and middle names
- aliases and transliterations
- dates of birth and partial-date handling
- nationalities and residences
- employers and professional roles
- profile URLs and confirmed identifiers
- adverse-media, sanctions, PEP, ownership, litigation, and insolvency terms
- language, script, and jurisdiction-specific query variants

Add deduplication rules, source prioritisation, query budgets, provenance tracking, and controls that prevent unnecessary or excessive requests.

3.2 Historical screening access

Develop an authenticated case-management interface for retrieving and reviewing historical screening records stored in Cloudflare D1.

- Case ID and applicant-name search
- screening-date and state filters
- report and audit-bundle retrieval
- authorised export
- access logging and audit history
- role-based permissions
- retention, legal-hold, and deletion controls

Historical access should be separated from the public evaluation interface and protected by individual authentication and authorisation.

3.3 Applicant-level risk profile

Develop a transparent and explainable Low, Medium, or High applicant risk profile supported by documented factors.

- sanctions and PEP results
- candidate-match confidence and identity conflicts
- adverse-media findings
- nationality and residence context
- source coverage and source failures
- unresolved potential matches
- analyst-confirmed information
- documented weighting, thresholds, overrides, and rationale

The classification must show contributing factors and confidence, remain reviewable, and never be based solely on one country, one source, or one candidate result.

3.4 Source resilience and provider governance

Improve resilience for external services affected by rate limits, timeouts, or temporary unavailability.

- bounded retries with exponential backoff
- provider-specific timeout and circuit-breaker rules
- clearer source diagnostics and fallback handling
- request scheduling and legally appropriate caching
- health monitoring and operational alerts
- documented distinction between required and optional sources
- provider terms-of-service, licensing, and data-protection review

Fallback sources should not silently change the meaning or evidential weight of a result.

3.5 Identity evidence and workflow governance

Add controls that move the system from public-source discovery toward a governed KYC/EDD workflow.

- documentary identity and address evidence
- evidence attachment and source citation
- analyst notes, approvals, and escalation
- four-eyes review for high-risk decisions
- risk-policy versioning and change control
- quality assurance, calibration, and false-positive review

These controls require legal, compliance, privacy, and security review before production use.

4. Disclaimer and Analyst Review Requirement

This application is an evaluation prototype and is not intended to operate as a production-grade compliance decision engine, a documentary identity-verification service, legal advice, or a substitute for professional analyst judgement.

All results require review and verification against original and authoritative sources. A source failure or unavailable source is not treated as a clear or no-match result. A no-candidate result is not proof that the subject has no relevant records. A potential match does not constitute a confirmed match.

Country-risk information provides contextual policy input only and does not establish an overall applicant risk classification. The current MVP does not claim to provide a complete Low, Medium, or High applicant risk score.

The final compliance decision remains the responsibility of an appropriately authorised reviewer operating under the organisation's legal, privacy, risk, and compliance framework.

Executive summary

Present MVP	Demonstrates the proposed KYC/EDD OSINT workflow, bounded provider integration, source-aware reporting, deterministic auditing, and manual-review handling.
Known limitations	Source coverage, query combinatorics, historical case access, individual authentication, documentary verification, and applicant-level risk scoring remain incomplete.
Planned development	Requires broader source design, stronger data governance, historical case management, identity evidence, resilience controls, and an explainable risk framework.
Evaluator reminder Use Demo A and Demo B for ordinary evaluation. Run a live screening only when necessary and only with authorised data. Select Forget token when finished.	

5. Technologies, Services, APIs and Logical Service Components

This inventory describes the final MVP implementation and its evaluation configuration.

Architecture statement

The final MVP is deployed as one Cloudflare Worker application with static assets and modular server-side components. It is not a distributed architecture of independently deployed microservices. The “logical service components” listed below are modules and bounded responsibilities inside the single Worker deployment.

Application and runtime technologies

Category	Technology / service	Role in the MVP
Edge runtime	Cloudflare Workers	Hosts the HTTPS gateway, screening API, orchestration logic, security controls, and report response.
Static web delivery	Cloudflare Workers Static Assets	Serves the evaluator HTML, CSS, and browser JavaScript from the same deployment.
Database	Cloudflare D1	SQLite-compatible storage for canonical screening runs, audit bundles, retention timestamps, and the exact upstream-usage counter.
Edge protection	Cloudflare Workers Rate Limiting bindings	Separate authentication and authenticated-API burst controls.
Secrets	Cloudflare Worker Secrets	Stores the evaluation access token and OpenSanctions API key outside source code and public assets.
Operations	Workers Deployments and Versions	Provides versioned production deployment and rollback references.
Observability	Cloudflare Workers Logs / Observability	Provides privacy-minimal invocation and error visibility without custom personal-data logging.
Scheduling	Worker scheduled event / retention purge	Deletes expired screening records according to the configured retention policy.
Runtime language	JavaScript ES modules	Implements gateway, connectors, matching, policy, storage, and report construction.
Browser client	HTML5, CSS3, vanilla JavaScript	Provides data entry, demo controls, progress, report rendering, JSON export, and print-to-PDF.
Network protocol	HTTPS, Fetch API, JSON	Transports authenticated browser requests and structured gateway responses.
Integrity	SHA-256	Creates deterministic report hashes and integrity metadata.

Category	Technology / service	Role in the MVP
Data query language	SQL / SQLite dialect	Defines D1 migrations, inserts, exact counters, indexes, and retention deletion.

External APIs and policy sources

API / source	Final state	Purpose
OpenSanctions API	Active	Two bounded requests per live screening: sanctions/watchlists and PEP/RCA candidate retrieval, each using threshold 0.6.
GDELT DOC API	Active	Bounded public-news and adverse-media discovery. Results are discovery candidates and source failures remain visible.
Brave Search API connector	Implemented, disabled	Optional fallback connector retained in the codebase but disabled in the final evaluation configuration; no Brave credential is configured.
FATF public statements	Policy input	Used as source material for the internal Q2 2026 country-risk policy matrix.
ISO 3166-1 alpha-2 country codes	Reference standard	Normalises nationality and residence inputs across the 249-country policy dataset.

Development, test, and delivery tooling

- Node.js and npm for local scripts, package management, and execution.
- Node.js built-in test runner (node --test) for automated regression and security tests.
- Wrangler CLI 4.92.0 for local development, dry runs, D1 operations, secret rotation, deployment, and deployment-status checks.
- PowerShell and curl for Windows-based operational validation and health checks.
- Microsoft Word / DOCX and PDF export for the evaluator handoff document.
- Browser print CSS for generated screening-report PDF output.

Logical service components (“microservices” within the single Worker)

Logical component	Responsibility
Evaluator web client	Static browser interface for access-token entry, subject data entry, demos, live screening, report display, JSON export, and PDF printing.
Evaluation-token authentication gateway	Validates the X-Evaluation-Token header against the server-side secret and enforces the access-expiry date.
Request validation service	Enforces HTTPS-origin policy, JSON content type, body-size limits, strict request envelopes, allowed fields, and input constraints.
Rate-limit and quota service	Uses Cloudflare rate-limit bindings for bursts and a D1 counter for exact upstream-query consumption.
Screening orchestrator	Coordinates the bounded source requests, candidate processing, section states, recommended actions, storage, and response.

Logical component	Responsibility
OpenSanctions sanctions connector	Performs the sanctions/watchlist candidate request.
OpenSanctions PEP/RCA connector	Performs the political-exposure and related-close-associate candidate request.
GDELT discovery connector	Builds bounded name-and-topic queries and normalises public-news discovery results.
Brave Search fallback connector	Optional connector present in the codebase but disabled in the final deployment.
Identity matching and evidence-gating service	Normalises subject/candidate attributes, preserves hard conflicts, and prevents unrelated candidate attributes from mutating the subject baseline.
Country-risk policy engine	Assesses analyst-provided nationality and residence separately and combines context according to the internal Q2 2026 policy.
Audit-bundle and report-integrity service	Builds the canonical ten-section audit bundle, source-execution audit, case metadata, and deterministic SHA-256 report hash.
D1 screening store	Creates and inserts canonical screening records while exposing no public history-retrieval route.
Retention purge service	Deletes screening runs after the configured seven-day retention period.
Synthetic demo fixture service	Produces deterministic Demo A and Demo B results with no external requests and no D1 screening storage.
Minimal health service	Provides GET /health with only {"status":"ok"} and makes no provider or D1 call.